Support Vector Machines

Instructions: Select exactly one option for each question. Unless otherwise stated, binary labels are $y_i \in \{-1, +1\}$, the intercept b is unregularized, and $C > 0$.
For questions concerning the standard soft-margin SVM, use

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i \quad \text{subject to} \quad y_i\big(w^\top \phi(x_i) + b\big) \geq 1 - \xi_i, \qquad \xi_i \geq 0.$$

Write $f(x) = w^\top \phi(x) + b$.

1. A linear SVM has

$$w = (3, 4)^\top, \qquad b = -10.$$

What is the Euclidean distance between the two canonical margin hyperplanes $w^\top x + b = +1$ and $w^\top x + b = -1$?

(A) $\frac{1}{5}$
(B) $\frac{2}{5}$
(C) 5
(D) 10

2. Consider the one-dimensional hard-margin training set

$$(-2, -1), \quad (-1, -1), \quad (2, +1), \quad (3, +1),$$

where each pair denotes (x_i, y_i). Which pair (w, b) solves the canonical hard-margin problem?

(A) $\big(\frac{1}{2}, 0\big)$
(B) $\big(1, -\frac{1}{2}\big)$
(C) $\big(\frac{2}{3}, -\frac{1}{3}\big)$
(D) $\big(\frac{1}{3}, -\frac{2}{3}\big)$

3. For a trained binary classifier, let

$$m_i = y_i f(x_i).$$

Which expression gives the signed geometric margin of training observation i when $w \neq 0$?

(A) $\frac{m_i}{\|w\|}$
(B) $m_i\|w\|$
(C) $\frac{|m_i|}{\|w\|^2}$
(D) $\frac{f(x_i)}{\|w\|}$

4. A decision function is replaced by

$$\widetilde{f}(x) = a f(x), \qquad a > 0.$$

Which statement is correct?

(A) Both predicted labels and geometric margins change.
(B) Predicted labels change, but geometric margins remain unchanged.
(C) Functional margins remain unchanged, but geometric margins change.

(D) Predicted labels and geometric margins remain unchanged, but functional margins scale by a.

5. At an optimum of the standard soft-margin SVM, a training point satisfies $0 < \alpha_i < C$. Which condition must hold?

(A) $y_i f(x_i) > 1$

(B) $y_i f(x_i) = 1$

(C) $y_i f(x_i) = 0$

(D) $y_i f(x_i) < 0$

6. At an optimum, a training point satisfies $\alpha_i = C$. Which conclusion is necessarily valid?

(A) The point is misclassified.

(B) The point lies exactly on the decision boundary.

(C) $y_i f(x_i) \leq 1$.

(D) $y_i f(x_i) = 1$.

7. At an optimum, a training point satisfies $\alpha_i = 0$. Which conclusion is necessarily valid?

(A) $y_i f(x_i) \geq 1$

(B) $y_i f(x_i) = 1$

(C) $y_i f(x_i) < 1$

(D) $y_i f(x_i) < 0$

8. A positive training observation has $f(x_i) = -0.4$. What is its smallest feasible slack variable?

(A) 0

(B) 0.4

(C) 0.6

(D) 1.4

9. A training observation has signed functional margin $y_i f(x_i) = 0.3$. How should this observation be interpreted?

(A) It is misclassified and has zero hinge loss.

(B) It is correctly classified but lies inside the margin.

(C) It lies outside the margin and has positive hinge loss.

(D) It lies exactly on the decision boundary.

10. Consider the hinge-loss objective

$$J(w, b) = \frac{1}{2}\|w\|^2 + C \sum_i \max\{0, 1 - y_i(w^\top x_i + b)\}.$$

Assume no training observation has margin exactly 1. What is its gradient with respect to w?

(A) $w + C \sum_{i:y_i f(x_i)<1} y_i x_i$

(B) $-C \sum_{i:y_i f(x_i)>1} y_i x_i$

(C) $w - C\sum_{i:y_i f(x_i)<1} y_i x_i$

(D) $w - C\sum_{i:y_i f(x_i)>1} y_i x_i$

11. The standard soft-margin SVM dual contains

$$\sum_i \alpha_i y_i = 0.$$

Which primal property produces this equality?

(A) The intercept is an unconstrained, unregularized variable.

(B) Every feature has zero mean.

(C) The kernel matrix has full rank.

(D) The slack variables are nonnegative.

12. Suppose a linear SVM is formulated with b fixed to zero. Which change occurs in its dual?

(A) The box constraints disappear.

(B) All multipliers must be equal.

(C) The quadratic term disappears.

(D) The equality $\sum_i \alpha_i y_i = 0$ is no longer required.

13. For a binary kernel SVM, which expression reconstructs the optimal weight vector in feature space?

(A) $w = \sum_i \alpha_i \phi(x_i)$

(B) $w = \sum_i \alpha_i y_i \phi(x_i)$

(C) $w = \sum_i C y_i \phi(x_i)$

(D) $w = \sum_i \alpha_i y_i x_i$ for every kernel

14. A trained kernel SVM has multipliers α_i. What is the squared norm of its feature-space weight vector?

(A) $\sum_i \alpha_i^2$

(B) $\sum_{i,j} \alpha_i \alpha_j K(x_i, x_j)$

(C) $\sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j)$

(D) $\sum_i \alpha_i K(x_i, x_i)$

15. A training set is linearly separable. As C increases sufficiently in the standard soft-margin SVM, what behavior can occur?

(A) The solution coincides with a hard-margin optimum.

(B) The optimal weight vector necessarily approaches zero.

(C) The classifier must eventually misclassify every point.

(D) The dual multipliers must all equal C.

16. Let $C_2 > C_1$. Compare globally optimal solutions of the same standard soft-margin problem at these two values. Which statement is guaranteed?

(A) Training misclassification count strictly decreases.

(B) The number of support vectors strictly decreases.

(C) The geometric margin strictly increases.

(D) The total optimal slack cannot increase.

17. Every training observation is duplicated exactly k times. The objective uses the *sum* of slack variables. Which new penalty parameter preserves the original primal objective after grouping identical copies?

(A) kC

(B) $\frac{C}{k}$

(C) k^2C

(D) $\frac{C}{k^2}$

18. Two implementations use

$$J_A = \frac{1}{2}\|w\|^2 + C\sum_i \ell_i, \qquad J_B = \frac{\lambda}{2}\|w\|^2 + \frac{1}{n}\sum_i \ell_i.$$

Which relationship makes their objectives equivalent up to multiplication by a positive constant?

(A) $C = n\lambda$

(B) $C = \frac{\lambda}{n}$

(C) $C = \frac{1}{n\lambda}$

(D) $C = \frac{n}{\lambda}$

19. All input vectors of a linear SVM are rescaled to $x_i' = ax_i$, where $a > 0$. Which penalty parameter C' makes training on the rescaled data equivalent to the original problem under $w' = w/a$ and $b' = b$?

(A) $C' = \frac{C}{a^2}$

(B) $C' = aC$

(C) $C' = a^2C$

(D) $C' = C$

20. A kernel is multiplied by a positive constant: $K' = sK$, where $s > 0$. Which choice of C' preserves an equivalent SVM decision function through the dual transformation $\alpha_i' = \alpha_i/s$?

(A) $C' = s^2C$

(B) $C' = sC$

(C) $C' = \frac{C}{s^2}$

(D) $C' = \frac{C}{s}$

21. A symmetric similarity function produces the Gram matrix

$$G = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

on two observations. What can be concluded?

(A) It is a valid kernel because all entries are positive.

(B) It cannot be a positive-semidefinite kernel on the stated domain.

(C) It is a valid kernel because the diagonal entries are 1.

(D) Its validity depends only on the class labels.

22. Let K_1 and K_2 be positive-semidefinite kernels on the same domain. Which construction is guaranteed to remain a positive-semidefinite kernel?

(A) $K_1 - K_2$

(B) $\frac{K_1}{K_2}$ wherever the denominator is nonzero

(C) $K_1 K_2$, evaluated pointwise

(D) $-K_1 - K_2$

23. Consider the homogeneous polynomial kernel

$$K(x, z) = (x^\top z)^3, \qquad x, z \in \mathbb{R}^2.$$

What is the dimension of its standard monomial feature map, using only monomials of total degree 3?

(A) 4

(B) 3

(C) 6

(D) 8

24. For $x, z \in \mathbb{R}^d$, consider

$$K(x, z) = (1 + x^\top z)^p,$$

where p is a positive integer. How many coordinates are required in the standard monomial feature map containing all degrees from 0 through p?

(A) d^p

(B) $\binom{d+p-1}{p}$

(C) $dp + 1$

(D) $\binom{d+p}{p}$

25. For the Gaussian RBF kernel

$$K(x, z) = \exp(-\gamma \|x - z\|^2),$$

what happens to the Gram matrix of a finite set of pairwise distinct observations as $\gamma \to \infty$?

(A) It approaches the all-ones matrix.

(B) It approaches the identity matrix.

(C) It approaches the zero matrix.

(D) It approaches the linear-kernel Gram matrix.

26. For the same RBF kernel, what happens to its Gram matrix as $\gamma \to 0^+$?

(A) It approaches the identity matrix.

(B) It approaches the zero matrix.

(C) It approaches the all-ones matrix.

(D) Its entries become negative.

27. Inputs for an RBF SVM are rescaled from x to ax, where $a > 0$. Which new value γ' preserves all kernel evaluations?

(A) $\gamma' = \frac{\gamma}{a^2}$

(B) $\gamma' = a\gamma$

(C) $\gamma' = a^2\gamma$

(D) $\gamma' = \frac{\gamma}{a}$

28. Assume an RBF SVM has finitely many support vectors. For a test sequence x_t whose distance from every support vector tends to infinity, what happens to $f(x_t)$?

(A) It necessarily tends to $+\infty$.

(B) It necessarily tends to $-\infty$.

(C) It necessarily tends to 0, regardless of b.

(D) It tends to b.

29. Which claim about the sigmoid similarity

$$K(x, z) = \tanh(ax^\top z + r)$$

is correct?

(A) It is positive semidefinite for all real a and r.

(B) It is not guaranteed to be positive semidefinite for arbitrary parameters and inputs.

(C) Symmetry alone guarantees its kernel validity.

(D) Bounded output alone guarantees its kernel validity.

30. Let K be a positive-semidefinite kernel satisfying $K(x, x) > 0$ for every input. Which statement about

$$\widetilde{K}(x, z) = \frac{K(x, z)}{\sqrt{K(x, x)K(z, z)}}$$

is correct?

(A) It is always indefinite.

(B) It is valid only when K is linear.

(C) It is positive semidefinite and has unit diagonal.

(D) It makes all distinct inputs orthogonal.

31. A kernel matrix K is centered using

$$H = I - \frac{1}{n}\mathbf{1}\mathbf{1}^\top.$$

Which matrix represents inner products between the centered training feature vectors?

(A) HKH

(B) $K - H$

(C) $H + K + H$

(D) $K^{-1}HK^{-1}$

32. Consider the XOR observations

$$(1,1),(-1,-1) \text{ with label } +1,$$

$$(1,-1),(-1,1) \text{ with label } -1.$$

Which feature is sufficient by itself to make these observations linearly separable?

(A) $x_1 + x_2$

(B) $x_1^2 + x_2^2$

(C) $|x_1| + |x_2|$

(D) $x_1 x_2$

33. Two training observations have identical inputs but opposite labels. For the standard soft-margin constraints, what is the smallest possible sum of their slack variables?

(A) 0

(B) 2

(C) 1

(D) 4

34. A hard-margin optimum is known, and one training observation satisfies

$$y_i f(x_i) > 1.$$

If only this observation is removed, which statement is valid?

(A) The optimal objective must strictly decrease.

(B) The remaining observations must become inseparable.

(C) The original optimum remains optimal.

(D) Every remaining observation becomes a support vector.

35. For a feasible linear hard-margin SVM, which uniqueness statement is generally correct?

(A) The optimal w is unique, while dual multipliers need not be unique.

(B) Every dual multiplier is unique, while w need not be.

(C) Neither w nor the dual multipliers can be unique.

(D) Strict convexity in w guarantees uniqueness of every primal and dual variable.

36. Every multiplier of a trained soft-margin SVM is either 0 or C; no multiplier lies strictly between them. What is the main implication for recovering b?

(A) The model must use $b = 0$.

(B) The model must be discarded as infeasible.

(C) One may use the usual free-support-vector equality with any training observation.

(D) One must use KKT bounds or another valid method because no free support vector is available.

37. A cost-sensitive SVM penalizes slack using

$$\sum_i C_i \xi_i, \qquad C_i > 0.$$

What are the corresponding dual box constraints?

(A) $0 \le \alpha_i \le \sum_j C_j$
(B) $0 \le \alpha_i \le C_i$
(C) $0 \le \alpha_i \le 1/C_i$
(D) $\alpha_i = C_i$ for every observation

38. False negatives are substantially more costly than false positives. The positive class is assigned a larger slack penalty than the negative class. Which statement best describes this change?

(A) It guarantees zero false negatives.
(B) It directly converts SVM scores into probabilities.
(C) It makes violations involving positive observations more expensive during optimization.
(D) It removes the need to validate the decision threshold.

39. A researcher standardizes all features using the entire dataset and then performs cross-validation. What is the principal methodological issue?

(A) Validation-fold information influences the fitted preprocessing transformation.
(B) Standardization makes every kernel invalid.
(C) Standardization forces all multipliers to be equal.
(D) Cross-validation requires unscaled features.

40. A researcher tunes C and γ using cross-validation and reports the best score from that same search as the final estimate of generalization performance. Which procedure better controls selection optimism?

(A) Increase both C and γ after tuning.
(B) Report training accuracy instead.
(C) Select the model with the most support vectors.
(D) Use nested cross-validation with tuning confined to the inner folds.

41. A multiclass SVM uses one-versus-one classification for K classes. How many binary classifiers are trained?

(A) K
(B) $\frac{K(K-1)}{2}$
(C) K^2
(D) $2K - 1$

42. Three one-versus-one classifiers produce: class A defeats B, class B defeats C, and class C defeats A. What follows under simple vote counting?

(A) Class A necessarily wins.

(B) Class B necessarily wins.

(C) All three classes receive one vote, requiring a tie-breaking rule.

(D) Such predictions are mathematically impossible.

43. A binary SVM outputs $f(x) = 2.8$ for a test observation. Which statement is justified without probability calibration?

(A) The score supports the positive class, but it does not directly specify a class probability.

(B) The positive-class probability is exactly 0.928.

(C) The positive-class probability is exactly 2.8.

(D) The observation is guaranteed to be correctly classified.

44. A model uses Platt scaling to estimate probabilities from SVM decision scores. Which description is most accurate?

(A) It replaces the SVM kernel with a polynomial kernel.

(B) It projects all support vectors onto the margin.

(C) It forces the SVM training objective to become logistic loss.

(D) It fits a sigmoid mapping from decision scores to estimated probabilities.

45. An RBF SVM has m support vectors, and each kernel evaluation for a dense d-dimensional input costs $O(d)$. What is the usual cost of computing one decision score, excluding unrelated overhead?

(A) $O(d)$, independent of m

(B) $O(md)$

(C) $O(m^3)$

(D) $O(d^3)$

46. A linear SVM is initially represented by

$$f(x) = \sum_{i \in SV} \alpha_i y_i x_i^\top x + b.$$

How can prediction be accelerated without changing $f(x)$?

(A) Remove every support vector with $\alpha_i = C$.

(B) Replace all coefficients by their mean.

(C) Precompute $w = \sum_{i \in SV} \alpha_i y_i x_i$ and evaluate $w^\top x + b$.

(D) Replace all inner products by RBF evaluations.

47. Consider squared-slack SVM training:

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + \frac{C}{2}\sum_i \xi_i^2$$

with the usual soft-margin constraints and $\xi_i \geq 0$. Relative to the standard linear-slack dual, which structural change occurs?

(A) The upper bounds $\alpha_i \le C$ disappear, and a quadratic penalty on α appears.

(B) Nonnegativity constraints on α_i disappear.

(C) The equality involving labels disappears despite an unregularized intercept.

(D) The kernel-dependent quadratic term disappears.

48. For ε-insensitive support vector regression, let the residual be $r = y - f(x)$. If $\varepsilon = 0.5$ and $r = -1.2$, what is the ε-insensitive loss?

(A) 0

(B) 1.2

(C) 1.7

(D) 0.7

49. In standard ε-SVR, a training observation satisfies

$$|y_i - f(x_i)| < \varepsilon.$$

What must hold for its two associated dual multipliers α_i and α_i^* at an optimum?

(A) $\alpha_i = \alpha_i^* = C$

(B) $\alpha_i = \alpha_i^* = 0$

(C) $\alpha_i + \alpha_i^* = C$

(D) $\alpha_i - \alpha_i^* = C$

50. Consider the one-class SVM formulation

$$\min_{w,\rho,\xi} \frac{1}{2}\|w\|^2 + \frac{1}{\nu n}\sum_i \xi_i - \rho$$

subject to

$$w^\top \phi(x_i) \ge \rho - \xi_i, \qquad \xi_i \ge 0, \qquad 0 < \nu \le 1.$$

Which pair of statements follows from its standard optimality conditions? Here, a training outlier means an observation with strictly positive optimal slack.

(A) ν is a lower bound on the outlier fraction and an upper bound on the support-vector fraction.

(B) ν fixes both fractions to exactly ν.

(C) ν is an upper bound on the outlier fraction and a lower bound on the support-vector fraction.

(D) ν affects only the intercept and imposes no constraints on either fraction.